7 **(a)** An electron is confined inside a one-dimensional potential well of length 0.50 nm with infinitely high potential barriers at $x = 0$ and $x = 0.50$ nm.

Within the well, the potential energy $U(x) = 0$, and outside the well, $U(x) = \infty$.

(i) By considering the de Broglie wavelength of the electron trapped in this well, explain why the electron has specific allowed energy levels.

The wavefunction of the electron must be a standing wave due to the infinite potential boundaries, thus its de Broglie wavelength can have only certain values. [B1] As momentum determines wavelength and kinetic energy is dependent on momentum, [B1] the kinetic energy of the electron can have only certain values.

[2]

(ii) The electron emits a single photon when it transitions from the second excited state to the ground state.

Determine the wavelength of the photon.

$$\Delta E = \frac{h^2}{8mL^2}\left(3^2 - 1^2\right) = \frac{\left(6.63\times10^{-34}\right)^2}{8\times9.11\times10^{-31}\times\left(0.50\times10^{-9}\right)^2}\left(3^2 - 1^2\right) \text{ [B1]} = 1.93\times10^{-18} \text{ J}$$

$$= \frac{hc}{\lambda} = \frac{6.63\times10^{-34}\times3.00\times10^{8}}{\lambda} \text{ [B1]}$$

$\lambda = 1.03\times10^{-7}$ m [A1]

wavelength = ... m [3]

(iii) Explain, with reference to the Heisenberg Uncertainty Principle, why the ground state energy of the electron cannot be zero.

As the electron is confined to the well, the maximum uncertainty in its displacement is L. [B1] Thus by the Heisenberg Uncertainty Principle, the minimum uncertainty in its momentum cannot be zero. [B1] Since there must be at least some uncertainty in its momentum, the electron cannot be at rest. NOT $E = pc = hc/\lambda$. Ignore any symbols if not defined clearly. [2]

(b) An electron bound within a hydrogen atom has allowed energy levels shown in Fig. 7.1.

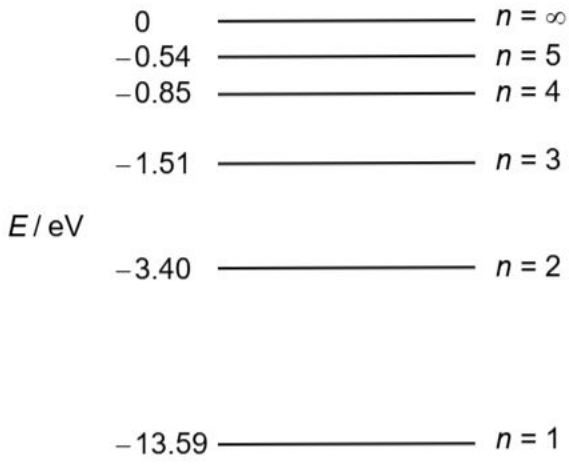


Fig. 7.1

(i) Suggest one reason why the energy levels of this electron are different from those of the electron in **(a)**.

The electron of a hydrogen atom has potential energy [B1] due to the electrostatic forces by the nucleus.

.....[1]

(ii) The electron emits a single photon when it transitions from the second excited state to the ground state. Determine the wavelength of the photon.

$\Delta E = (-1.51 - (-13.59)) = 12.08$ eV $= 1.933 \times 10^{-18}$ J [B1] ecf from previous part if used $n = 2$

$$= \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{\lambda}$$

$\lambda = 1.029 \times 10^{-7}$ m [A1]

wavelength = ... m [2]

[Total: 10]